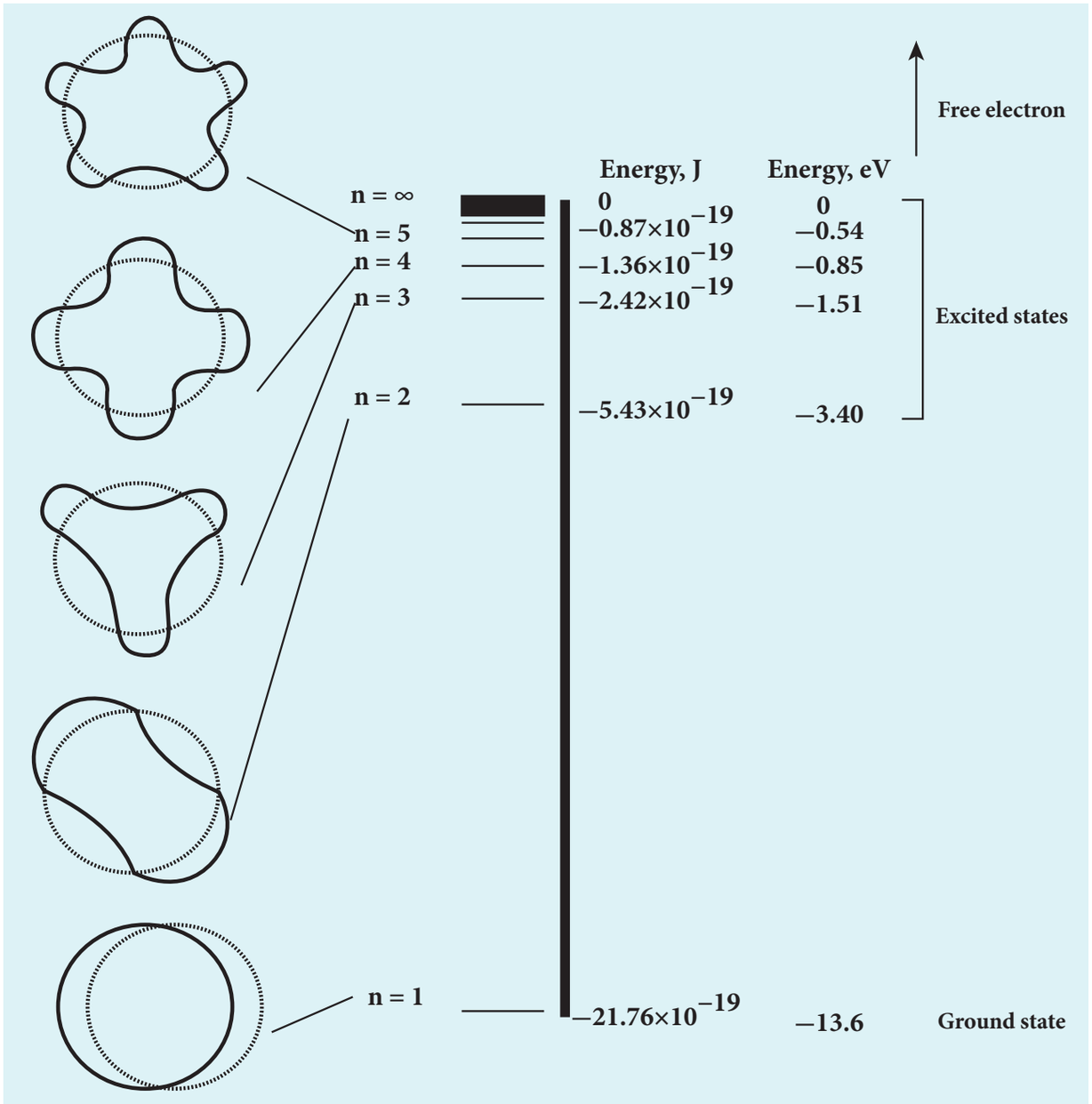


Figure 9.20 **Energy levels of a hydrogen atom**

$$\upsilon_n = \frac{e^2}{2\varepsilon_0 h}\frac{Z}{n} = c\left(\frac{e^2}{2\varepsilon_0 hc}\right)\frac{Z}{n} = \frac{\alpha cZ}{n}$$

where c is the speed of light in free space or vacuum and its value is $c = 3 \times 10^8$ m s^{-1}and α is called fine structure constant.

For a hydrogen atom, $Z = 1$ and for the first orbit, $n = 1$, the ratio of velocity of electron in first orbit to the speed of light in vacuum or free space is

$$\alpha = \frac{(1.6\times10^{-19}\,\text{C})^2}{2\times(8.854\times10^{-12}\,\text{C}^2\text{N}^{-1}\text{m}^{-2})} \times \frac{1}{(6.6\times10^{-34}\,\text{Nms})\times(3\times10^8\,\text{ms}^{-1})}$$

$$\frac{\upsilon_1}{c} = \alpha = \frac{e^2}{2\varepsilon_0 hc}$$

$$\approx \frac{1}{136.9} = \frac{1}{137}$$ which is a dimensionless number

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